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A Pedagogic Pattern Model for Up-Skilling Software Engineering Students in HCI Design Practice

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Abstract. In this chapter, we describe a pedagogical pattern process that structures and transcribes salient points of simulation-based learning applied to Human-Computer Interface (HCI) design. First, we outline Christopher Alexander's pattern language design theory. Second, we examine Jean Houssaye's trialectic Pedagogic Triangle. Then, we combine key aspects of Alexander's and Houssaye's theories in order to document instructional practices in the teaching of engineering design. The practical and theoretical relevance of our unique approach is summarized in our Pedagogic Pattern Model (PPM). The Model examines the instructional practices of experienced university instructors involved in HCI design courses. Through collective analysis, PPM focusses on capturing reflective classroom experiences via structured case studies of teaching HCI design practice.

Keywords: pedagogical design patterns; simulation-based education; interaction design; pedagogic triangle; instructional and design practices; Pedagogic Pattern Model (PPM)

1. INTRODUCTION

Design is a fundamental issue in software engineering, computer science as well as the management of Information Systems programs. This can be explained by the growing pressure on engineers, and other design-oriented professionals, to ground their design concepts and decisions on a systematic body of knowledge and empirical evidence of successful design practice (cf. Van Aken and Romme 2009). Hevner and Chatterjee (2010) refer to Design Science Research in Information Systems. They include a selection of papers from the Design Science Research in Information Systems and Technologies conferences (DESRIST) that look at key principles of Design Science Research, and the integration of research on design sciences linked to design practices. In this context, this chapter addresses the increased demand for more clearly relevant design-oriented research and training based on real-world problems.

However, few publications exist in formalizing the way in which engineers can be trained in design with regards to existing good practice, and to issues, topics, and paradigms that the field covers and those grounded in related domains. These domains include the study of the psychological and social aspects of users, their behavioral styles and patterns, user experience modeling, design and prototyping tools, user-oriented and usability evaluation, traditional and future design paradigms, as well as the role of theory in decision-making during design and design conception (e.g., Bergin, 2012; Pedagogical Patterns Project, 2001; Gamma, Helm, Johnson & Vlissides, 1995). One manifest challenge of these domains is the way in which good practice can be formally captured, documented and disseminated. More often than not, practices embedded in a specific project are difficult to reapply in other projects and

organizational contexts. In effect, little work and tools exist to guide practitioners in capturing their successful, and unsuccessful, design practices. A Finnish study confirms this when it points out that: "… Most of the teachers aimed at promoting purposeful inquiry in their pedagogical designs did not necessarily know good methods and practices for structuring and scaffolding students' inquiry efforts" (Lakkala, Lallimo, & Hakkarainen 2005 p. 351).

This observation is equally true in the teaching of design to software engineering students. These students are often not familiar with the underlying theories and models of design. For example, many software engineering teaching programs include, invariably, a course on software design with UML (Unified Modeling Language), software design patterns as well as a series of user interface design lectures. Such instructional programs are not enough to train software developers in critical design skills. In addition, the large majority of software and HCI design textbooks do not address instructional design good practice and how instructors need to, or should, present such good practice to students. Added to this, it is not uncommon for cognitive psychology-based user interface design to be seen as a nuisance that gets in the way of "real and traditional" software design. This results in the tendency of many software engineering students and instructors to consider user interface design practices and knowledge as haphazard and thus unimportant. This explains, for a large part, why students find it difficult to fully understand and master user interface design methods. Clearly, then, the knowledge and training given is not sufficiently appropriate. If we accept the fact that user interface design is indeed a key component in modern software development, the question is then how to teach such a challenging topic?

We argue that students and teachers need to go beyond ad hoc descriptions of design practices by examining how design patterns can be scripted, justified and used via current design tools and technics. What is needed is a robust instructional approach for design education. In this sense, the essential features of patterns-based good design practice need to be logically and explicitly coherent from an instructional point of view. It is in knowing the logical underpinning of how an instructional practice can be captured and documented in order to empower instructors to share good design practice from one context to another in terms of learners', and other stakeholders', needs. In this case, it is in adopting a theory-driven logic to teaching/learning (pedagogics) that instructors can be empowered not to blindly reproduce someone else's practice at the expense of the unique possibilities of their own context. We argue that it is through the understanding of the theoretical (epistemological) dynamics of a teaching/learning process that can lead to improving instructional practice.

With this in mind, we propose a Pedagogic Pattern Model (PPM) as a framework to communicate the salient points of simulation-based learning when designing HCI concepts and practices. PPM is related to the emerging notion of "pedagogical patterns", which has been introduced in different domains notably in computer science through the *Pedagogical Patterns Project* (Bergin, 2012). The aim of the pedagogical patterns approach is to capture the essence of practice that can be communicated to those who need the knowledge, such as new instructors needing to learn "what is known by senior faculty by easy transference of knowledge of teaching within the community" (Bergin, 2012). This is in line with Wade (2002) for whom "pedagogical design patterns" seeks to capture expert teaching practice, communicate expertise, solve common recurring problems, provide a vocabulary of solutions, and work with other patterns. Such as this approach is presented, it appears that these documented "patterns" are portrayed as inherently "pedagogical" in themselves (reminiscent of a type of Platonic essentialism).

In order to avoid epistemological, if not semantic confusion, the specificity of the Pedagogic Pattern Model is that the knowledge creation process resides in underlying socio-

cultural practice mediated by computer based interaction, and is not intrinsically in what can be identified post hoc as discernible "patterns". In this sense, the encapsulation of a given socio-cultural HCI practice – emerging as a designated "pattern" – represents one key epistemological element, amongst others, that needs to be taken into account in a knowledge creation process. In short, patterns act as a *starting point* to be adapted to the local socio-cultural HCI context, based on an explicit "model", i.e., a synthetic portrayal of the generic features of a pattern-building process. A set of documented patterns is thus not a manual of one size fits all, ready-made recipe to be applied blindly. It is vital to emphasise this point in order to avoid confusion about the role and limits of such the PPM approach.

In operational terms, PPM uses a unique type of "alphabet" to transcribe key instructional events. In doing this, the alphabet allows a coherent transposition of events from one given context to another. This transposition implies keeping the essential features of a sequence of events, while altering its more secondary features to the possibilities of a given context.

The originality of PPM is that it provides a guideline for structuring instructional frameworks in a logical manner based on good practice. First and foremost, we contend that the inner driving force of such an approach depends on the inventiveness that a teaching/learning framework affords the instructor. In this sense, French philosopher, Chartier (1932/1967) espouses this point of view in highlighting a paradox of inventiveness. For Chartier (1932/1967 p.106) the ingenuity of tomorrow comes from the study of tried and tested practices of the past. How then to identify and render coherent these past practices? One recognized authority in the world of science and engineering, who has adopted such a view of good practice, is design architect Christopher Alexander.

Alexander developed his approach after observing that users understand more about designing the building they need than a qualified expert-architect. The role of such an expert is to capture and harness users' lived knowledge. Based on this observation Alexander, Ishikawa, Silverstein, Fiksdahl-King & Angel (1977) produced an approach, called "pattern language" in designing and building different objects. This approach gave rise to the design pattern movement that has crossed over to other domains ranging from software design to the social sciences (e.g. Labour and Kolski 2010, pp. 126-129.

2. A FIVE-STEP APPROACH TO USING A PEDAGOGIC PATTERN MODEL

The five basic phases of the "Model" (in the sense of portraying essential elements and their relationships within a given structure, see Table 1 below) can be summarized in the following way (Labour and Kolski, 2010):

A. General description and context
B. Spatio-temporal boundary markers
C. Overall action plan of pedagogical techniques/tools
D. Group interactions at each pole of Housaye's Pedagogic Triangle
E. Feedback

A. General description and context

It is in this initial phase that the overall teaching/learning (pedagogic) situation is set. The aim is to describe what Alexander (1979, p.253) calls a "systems of forces" that bonds a series of opportunities and limits of a given situation. This description is visually represented by French educationalist, Houssaye (1992, p. 235, 1994) as an all-encompassing circle (Figure 1, see below).

B. Spatio-temporal boundary markers

Temporal markers act as markers of when a sequence formally starts and ends. In this case the *TØ* marker designates the *Initial state* of the instructional sequence for the learner, the instructor, and the researchers. The *T1* marker indicates the *"Final" state* of the instructional session for the learner and the instructor. The *T2* marker identifies the *"End" state* of the researcher's analysis of what was done and what could have been done in the instructional session.

Spatial markers pinpoint two basic levels of analysis regarding instructional sequences. Firstly, there is the "Classroom context" associated with instructor-student interactions in a given space-time. Secondly, the markers also indicate the "Discourse community context. In this way PPM takes into account, in the design of instructional sequences, that the curriculum of future engineers is bound by norms (e.g. of what is considered as legitimate knowledge) and values (e.g. deontological) of a scientific and professional community. Ideally, there should be a reciprocal relationship between the two levels through the description and analysis of the changing needs, preferences, and expectations of "Classroom context".

C. Overall action plan of pedagogical techniques/tools

In order to render the action plan operational, a diagrammatic system was developed based on Houssaye's Pedagogical Triangle (1992). In essence, the Pedagogical Triangle consists of three apexes – Knowledge domain, Instructor and Learner – circumscribed by a circle indicating the limits and the opportunities of a given instructional context (see above "General description and context"). Each apex interacts with another and this leads to formally distinct learning processes, see Figure 1 (below). Each process is visually represented by three triangles with different shaded sides, see Figure 2 below.

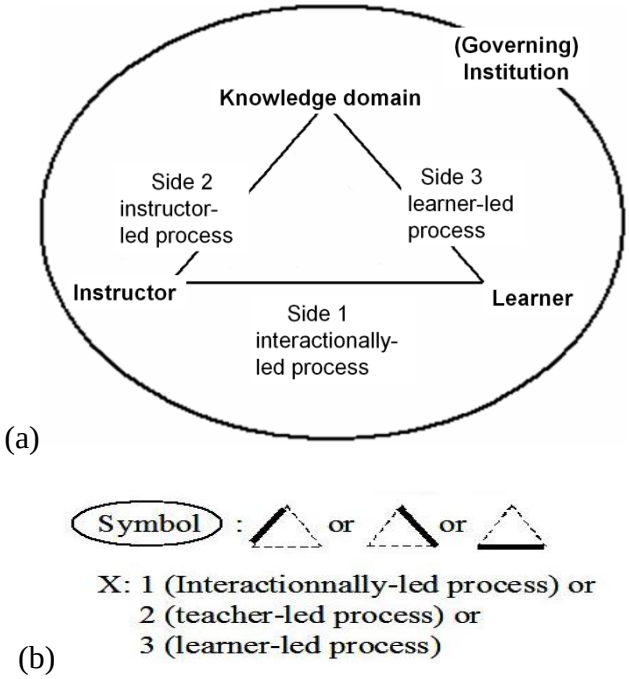


Figure 1. (a) The Pedagogic Triangle adapted from (Houssaye 1992), (b) Symbols indicating the three learning processes of Houssaye (1992)

In using the Pedagogic Triangle, a series of task boxes are proposed indicating which learning process predominates in a given sequence (Labour and Kolski 2010: 118, 120).

Houssaye argues that if more than one instructional process is "activated" simultaneously, there is likely to be cognitive overloading, if not conflict between processes that are intrinsically different. Each symbol is then put in a "Task box" with a description of the predominating learning process, according to the "action plan" and a succinct description of the task objective. The appropriate spatio-temporal boundary markers (TØ, T1, or T2) are found on the left hand side of the Task box, see Figure 2 below. There are no limits as to the number of boxes that can be put in the model. Task boxes are linked by arrows. Downward arrows show the initial flow of the sequences, and upward arrows indicate feedback dynamics.

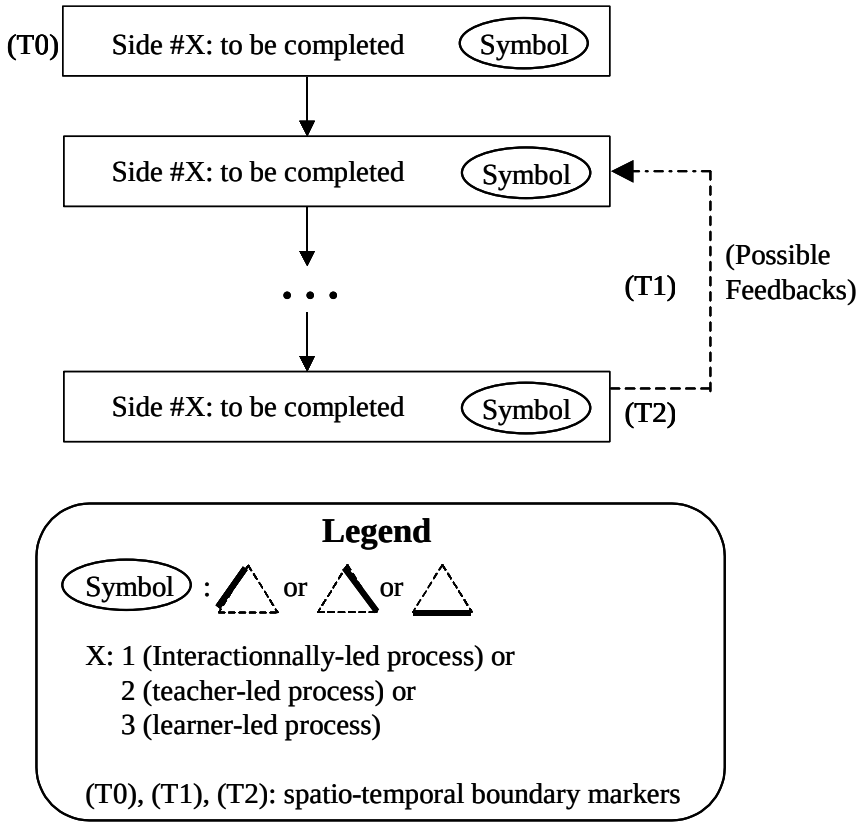


Figure 2. Guide to the process of the overall action plan

D. Group interactions between poles of the Pedagogic Triangle

The phase describes the selected presence of an interaction linked to an anticipated learning process. These interactions involve intra- and inter-group processes between the Learner-led, Instructor-led and knowledge domain poles of Houssaye's Pedagogic Triangle (see Figure 1 above). The interactions need to be regulated in order to avoid cognitive overloading or conflict between the processes. Such interactions represent what the ancient Greeks called *pharmakon*, i.e. certain phenomena are neither good nor bad, they can cure or kill, it all depends on their dosage.

E. Feedback

Systematic feedback is indispensable in order to adapt to the changing needs of learners and instructional contexts. The feedback reexamines the sequences and their effects on learners.

In the section that follows a specific instructional context is formalized according to the principles of the Pedagogic Pattern Model. To do this, we present a simulation-based learning context in an HCI design case study.

3. CASE STUDY IN HCI DESIGN

A. Protocol

Since the beginning of the 1990s, we have conducted various studies in several HCI design courses. We asked engineering students to play the role of HCI designers of interactive systems in rival "companies". The studies were conducted with classes of 20 to 25 students in their fourth year of a Bachelor's degree in Electrical Engineering and Industrial Computer Science (average age of students is 23 years), and with students doing a Master's degree in Information Technology (average age is 24 years).

The protocol used in our studies is made up of three phases. The first phase includes a preliminary preparation phase of a 12-hour instructor-directed class in HCI design. The application domain is focused on the design of adaptable, adaptive and personalized HCI as used by different types of users. The pedagogic content is grounded on several works, such as those of Rasmussen (1986), Schneider-Hufschmidt et al. (1993), Hoc and Amalberti (1995), Calvary et al. (2003), Brossard et al. (2011), as well as the instructor's personal professional experience gained in several industrial projects.

During the second phase of six hours (90 min a week over 4 weeks), the students were organized in teams of four or five. All the teams were seated in the same room, with each team being only several meters from each other. Each team represented a company. The companies (teams) competed against each other in submitting a tender for the design of an adaptable, adaptive and/or personalized HCI system to be used by several types of users in a complex industrial organization. The projected users were to perform tasks in normal and abnormal situations.

One of the rules of the simulation is that the teams should avoid revealing information to their "competitors". During the sessions, the teams were able to question the instructor who acted both as the game master and an "employee" made available to competing companies by the organization that had launched the invitation for tender. At the end of the week, each team had to submit a specification dossier (in PDF format by email) presenting their analysis of the problem and a model-based description of an adaptable, adaptive and/or personalized HCI. Three extracts of specification dossiers are shown in the following figures: (a) a UML use case of different types of users (see Figure 3 below), (b) two screen pages extracted from a mockup (see Figure 4a below), and (c) an UML sequence model describing the dynamics of a proposed interactive application (see Figure 4b below).

In the third phase, we collected feedback from the instructor about each student's dossier. We used 35 evaluation criteria deemed crucial in the HCI field (usability problems, relevance of the solution, methodological weaknesses, quality of the models used, …). A panel of instructors studied and graded the dossier of the teams and the oral presentation of their work.

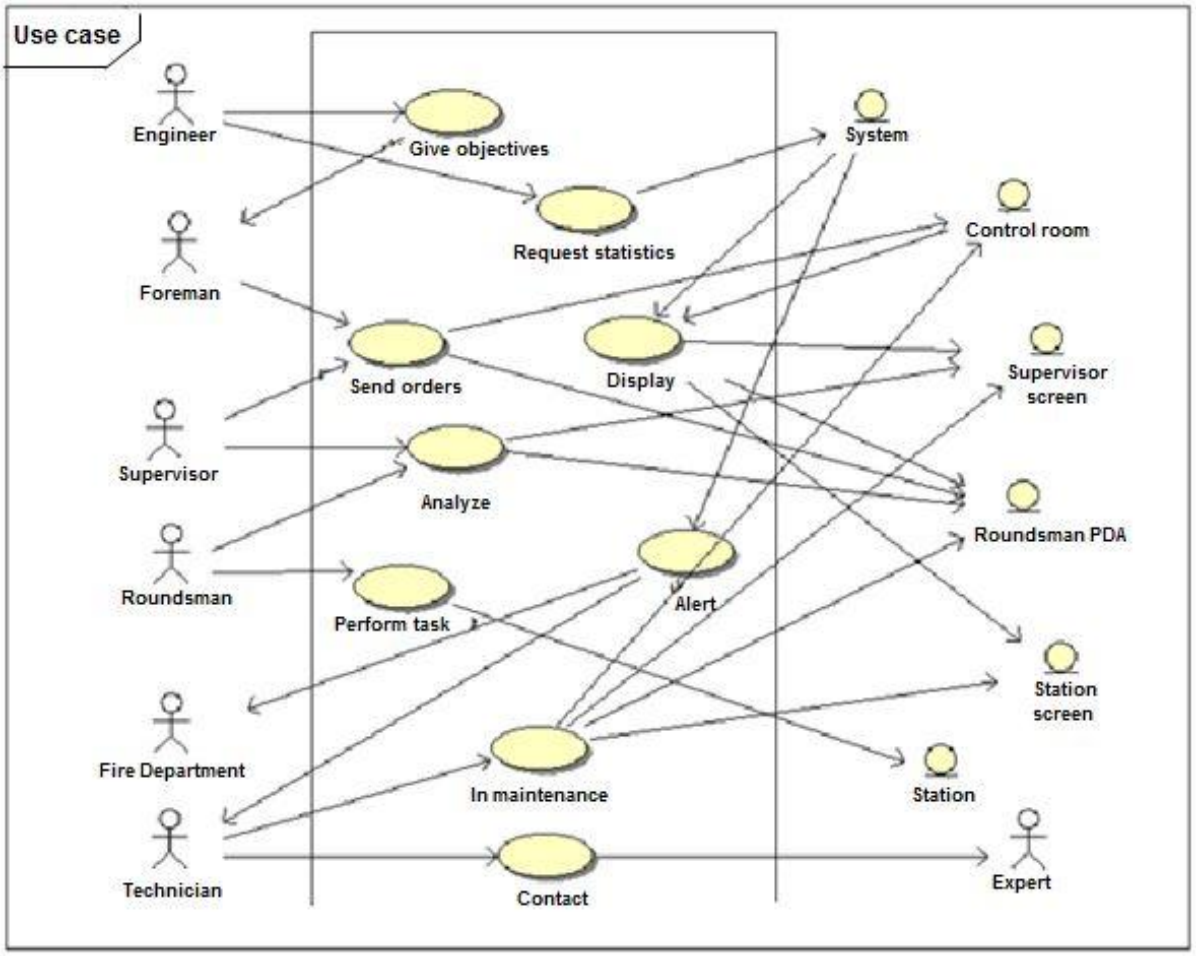


Figure 3. Example of an UML use case, showing seven types of users (given in a specification dossier)

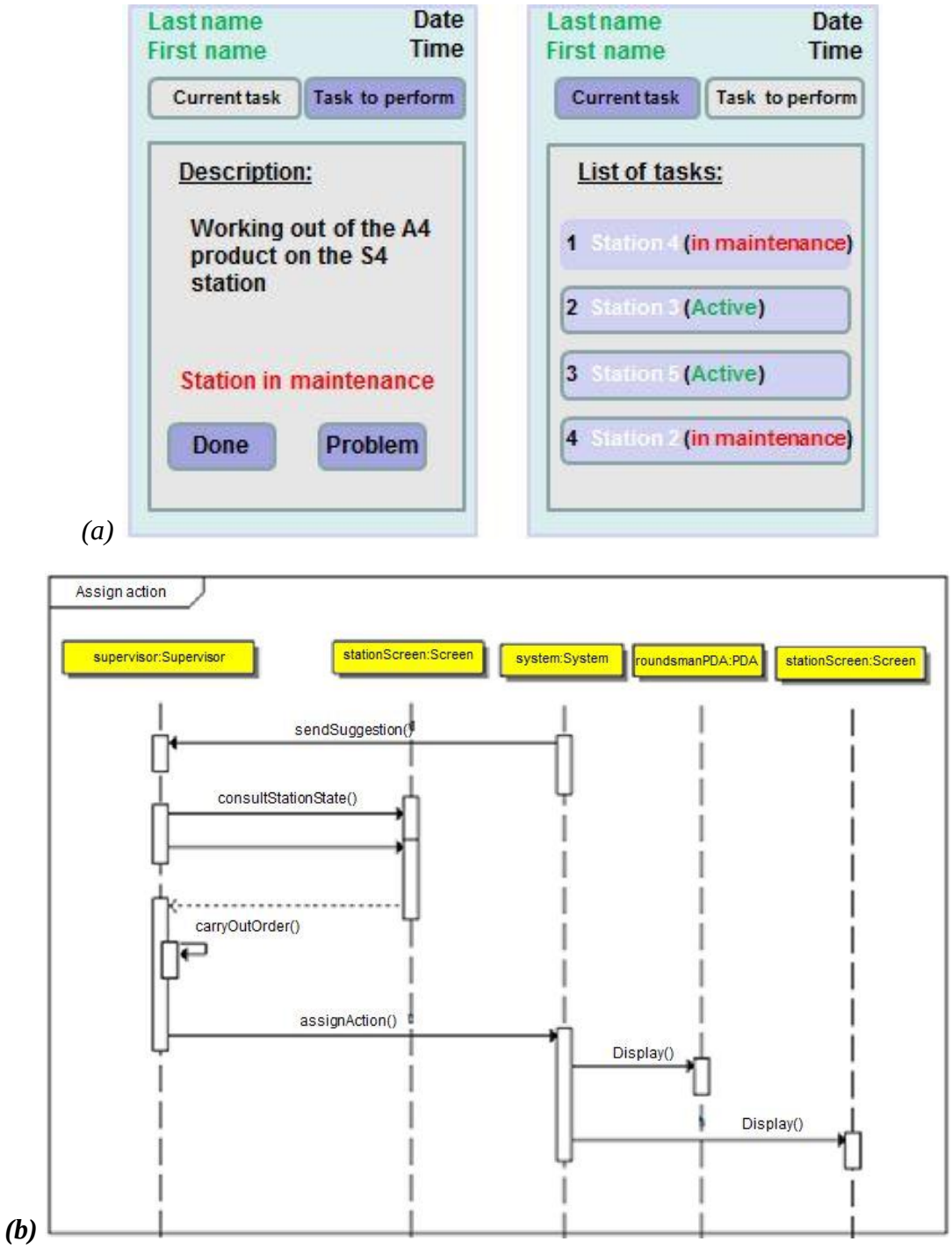


Figure 4. (a) Example of two screen pages, as part of a mockup of an interactive system available on a PDA (given in a specification dossier) (b) Example of an UML sequence diagram (given in a specification dossier) based on an order sent from one actor to another via the interactive system

B. Spatio-temporal boundary markers

At the heart of our model is section B. To characterize the experience found in the case study, we drew up Table 1 below. The table portrays a double-entry format juxtaposing the "classroom" (local) and "discourse community" (global) contexts in terms of temporal markers. This description is followed by a diagrammatic expression of the Pedagogic Pattern Model (Figure 5, see below).

A.1. Knowledge domain pole of the Pedagogic Triangle:		
	Classroom context	**Discourse community context**
Aims	Without object	To improve the identification of design errors of potential future designers
"Initial" state	**(at TØ)** Copy of the slides used and explanations given to the students	**(at TØ)** Existing knowledge including resources (e.g. books) and explanations given to the students
"Final" state	**(at T1)** The same as TØ plus learners' solutions and instructor's comments	**(at T2)** The same as TØ.
A.2. Instructor pole of the Pedagogic Triangle:		
	Classroom context	**Discourse community context**
Aims	To teach HCI design	To learn about design errors in HCI design
"Initial" state	**(at TØ)** Intention to encourage students to learn, *viz*. through simulation	**(at TØ)** Intention to acquire data concerning students as potential HCI designers
"Final" state	**(at T1)** Satisfaction that the students can design a HCI system	**(at T1)** Intention to analyze available data (i.e. the dossiers)
A.3. Learner pole of the Pedagogic Triangle::		
	Classroom context	**Discourse community context**
Aims	To pass the exam	To obtain the degree and get a job with skills required for the job market
"Initial" state	**(at TØ)** Acquiring skills to pass the exams	**(at TØ)** Hope to acquire knowledge about HCI design
"Final" state	**(at T1)** Satisfaction of developing skills *via* the production of a quality HCI specification dossier	**(at T1)** Confidence to be able to produce real HCI specification dossiers in companies (knowledge transfer)

Table 1. Pedagogic Pattern Model of Case study #1: Simulation of adaptable, adaptive and/or personalized HCI design

C. Action plan of pedagogical techniques or tools used

Taking account of the three-part rule of the design pattern approach, a diagrammatic representation of the action plan is presented in Figure 5 below. The figure portrays the overall process in a more schematic way, followed by a step-by-step commentary of the structure that our Pedagogic Pattern Model affords its user. In this way, Table 1 and Figure 5 present the data in a way that they can be compared and contrasted with other similar pedagogic contexts. These elements suggest that our approach is able to identify common instructional features in different contexts.

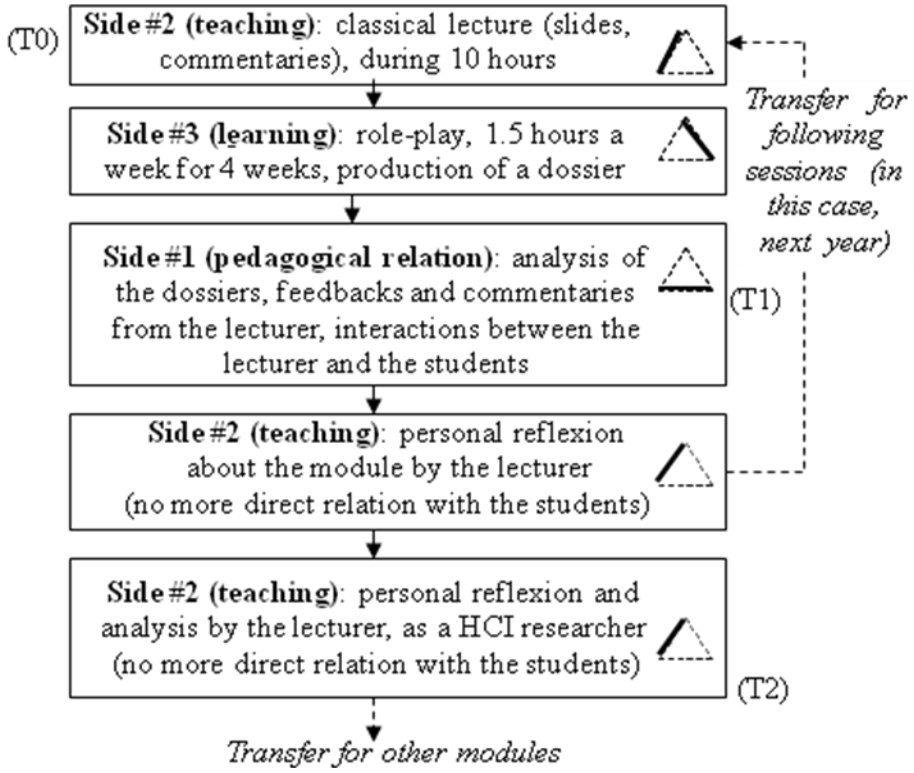


Figure 5. Overall action plan of an HCI design simulation

Table 1 shows an overall process that goes from TØ to T1 based on the three different phases of the Pedagogic Triangle. The first phase is an instructor-led process (side #2 of the Pedagogic Triangle). The second phase is a learned-led process (side #3 of the Triangle). The third is an interactionally-led process (side #1 of the Triangle). All three sides are concerned in the teaching/learning process, but at different given moments in order to avoid conflict between teaching/learning processes. This is followed by an instructor-led process (side #2 of the Triangle) during which the instructor undertakes a self-evaluation about the module. It is at this juncture that a loop back of information operates to improve future instruction sequences (see Figure 5 below). Finally, the instructor-led process (side #2) moves on to a broader analysis as a HCI researcher, within a given discourse community, to finish at T2. The consequences of the Pedagogic Pattern Model portrayed in Table 1 and in Figure 5 are made explicit under the concepts of "group interaction" (see Section D below) and "feedback" (see Section E below).

D. Group interactions at each pole of the Pedagogic Triangle

Interactions at the "learner" pole:

- **Interaction between groups of learners:** The sub-groups (teams) were in competition as rival "companies". It was therefore important, for each group, to avoid transmitting important data to the other groups.
- **Interaction within groups of learners:** As the work was difficult and long, members of each group had to cooperate to finish their task. The instructor noted down the exchanges of ideas and points of view of each group.
- **Interaction with the institution:** There was a small possibility that the students had discussed with students from previous years, or with other instructors (though few of

them were specialized in HCI) about the on-going project. This was a calculated risk taken by the instructor, but such intrusive influence has never been identified.

Interactions at the "instructor" pole:

- **Interaction between groups of instructors:** not applicable as the instructor worked alone.
- **Interaction within groups of instructors:** The instructor had discussed with colleagues about innovations in pedagogy, but there was no discernible consequence on the module in question as its action plan had been predefined, based on previous experiences of other modules.
- **Interaction with the institution:** The instructor took into account the fact that other classes had also conducted similar simulations involving HCI design and evaluation in the previous year, and software engineering in the two previous years.

Interactions at the "knowledge domain" pole:

- **Interaction between knowledge domains:** Along with pedagogical texts from the instructor, other texts were available to students from the Internet and the University library.
- **Interaction within a knowledge domain:** In certain groups, there were several students with a specialized knowledge in HCI (acquired in past projects or internships). The instructor noticed in such cases that these students attempted to share their experience with the group.
- **Interaction with the institution:** Other faculty staff could have helped students to improve the quality of their specification dossiers but students did not avail of this help.

E. Feedback

- **Feedback from the "learner" pole:** Each of the groups participated in the simulation with a high degree of motivation as witnessed by the virtual lack of absenteeism in the simulation classes, and by the fact that all the dossiers handed in included many positive aspects in accordance to the instructor's evaluation criteria. Thanks to the simulation, the potential designers felt they had worked on a project that was close to industrial reality. The students also appreciated the competitiveness between the groups. In short, there was no visible presence of learner boredom, or the intrusion of the all-knowing instructor interfering with learners' work.
- **Feedback from the "instructor" pole:** Learners appreciated the simulation as a valuable asset to their formal university instruction. The professional nature of student dossiers and their active participation satisfied the expectations of the instructor. During the oral examination a panel of instructors questioned the learners. This provided valuable input to the instructor in how to better explain the general description and context of the simulation module in the following years. This feedback leads to the conclusion that the danger of learners falling back on prefabricated knowledge was not a dominant feature of the simulation.
- **Feedback from the "knowledge domain" pole:** An analysis of the dossiers will be followed by the submission of a scientific publication as a way to get feedback from the HCI community.

4. CONCLUSION

This chapter investigates the underlying pedagogy of teaching design practices for software developers. The design practices are, or can be, captured in the format of HCI design patterns as defined in the other chapters of this book. We propose a Model to capture good instructional practice seeking to develop knowledge creating process in students. Our Pedagogic Pattern Model (PPM) is based on patterns that have been recognized in many areas of education involving group work, software design, and HCI. The approach can be seen as both an extension of HCI design patterns, and of pattern languages in general aimed at fostering the teaching of design via good practices.

More specifically, the chapter examines the experiences of instructors whose goal are to capture pedagogic patterns in teaching design. To do this, PPM marries key aspects of Christopher Alexander's pattern language approach to Jean Houssaye's trialectic theory. This led to a Pedagogic Pattern Model which consists of five phases: (1) General description and context, (2) Spatio-temporal boundary markers, (3) Overall action plan of pedagogical techniques or tools used, (4) Group interactions at each pole of the Pedagogic Triangle, and (4) Feedback.

The strength of the Model is that it presents different instructional constructs in an educationally coherent manner. This is achieved through a specially developed transcription mode, which describes events concerning the learner, instructor and knowledge domains. In this chapter, the transcription of instructional practice focuses on simulations in Human-Computer Interface design (HCI) by effectively incorporating the cognitive, emotional and socio-cultural aspects of users' experiences and behavior patterns.

It is hoped that the capacity of PPM to transcribe and document good practice contributes to a more hands-on interdisciplinary approach in the dissemination of design practice when preparing our students for the professional world. From this perspective, the Pedagogic Pattern Model contributes to an overall vision of HCI design as a domain at the intersection of computer sciences, design sciences, and social sciences, including the fields of education and psychology.

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